Correlation coefficient

Exercises/CI_and_correlation.ipynb

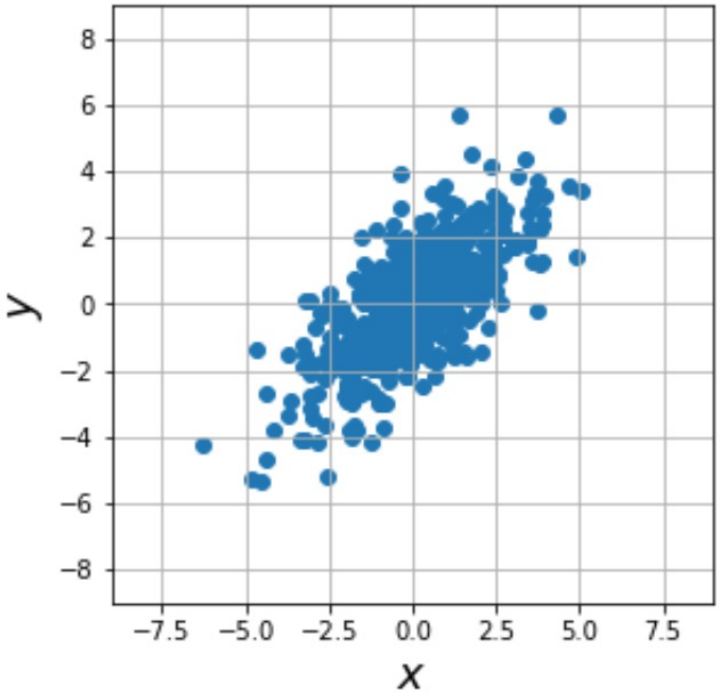
Pearson r_{xy} (linear) correlation coefficient

$$r_{xy} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2}}$$

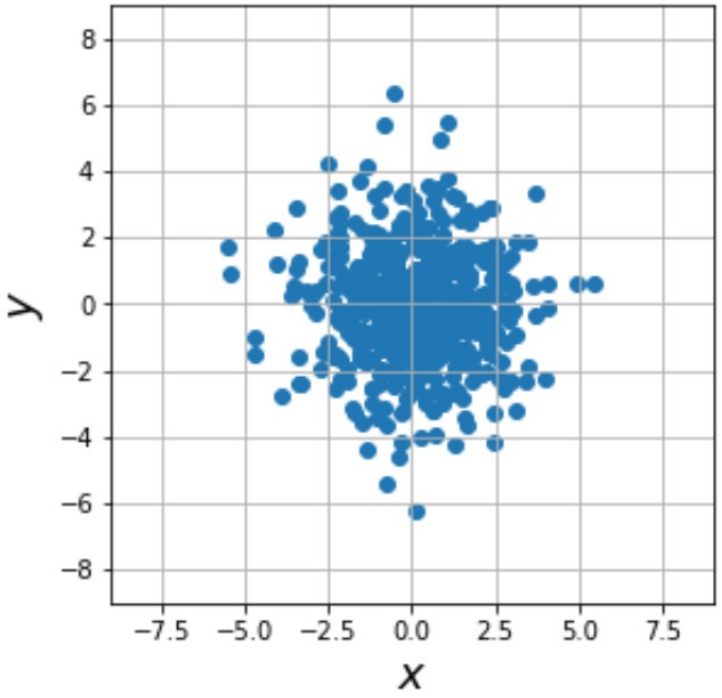
Quantifies existence of a **LINEAR** correlation between x and y

$-1 < r_{xy} < +1$

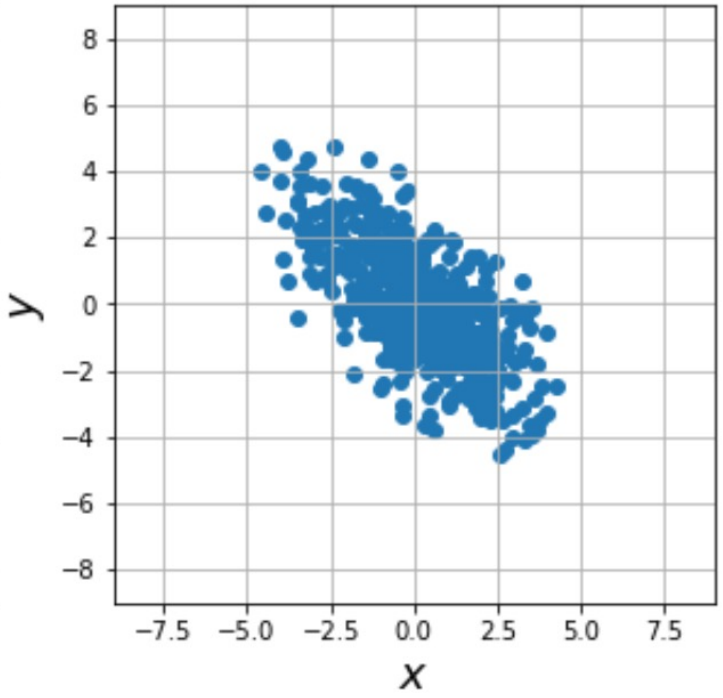
$r_{xy} > 0$



$r_{xy} \approx 0$



$r_{xy} < 0$



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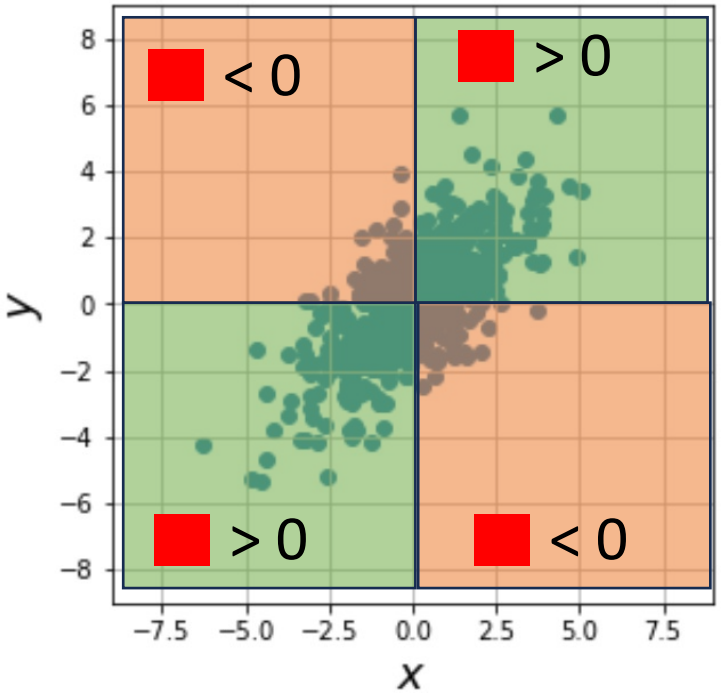

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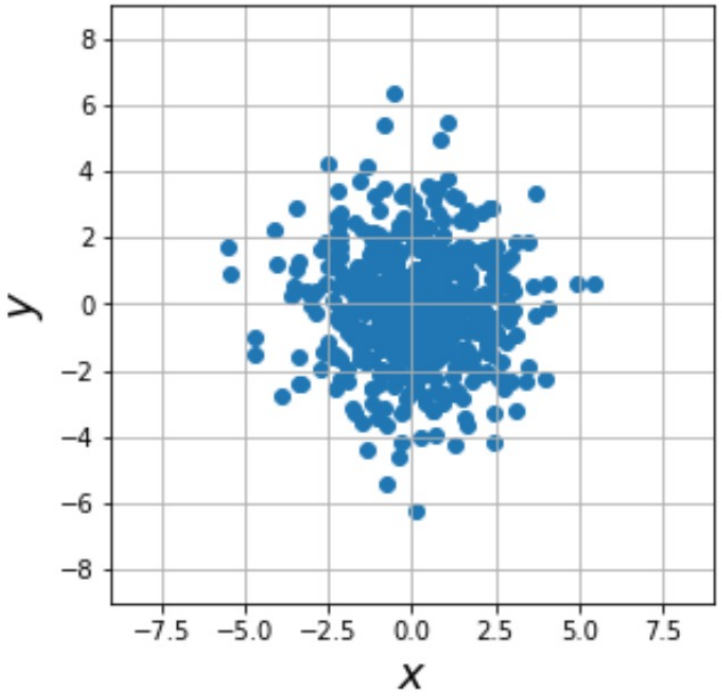

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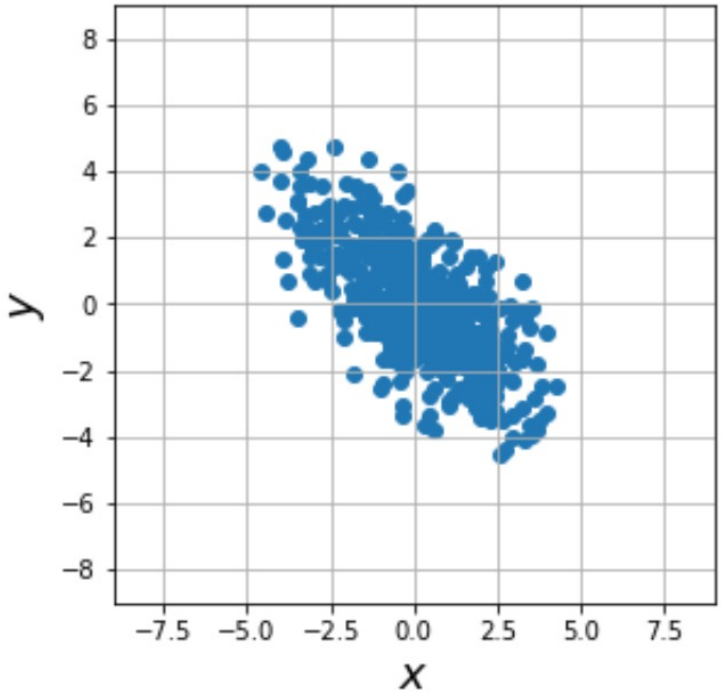

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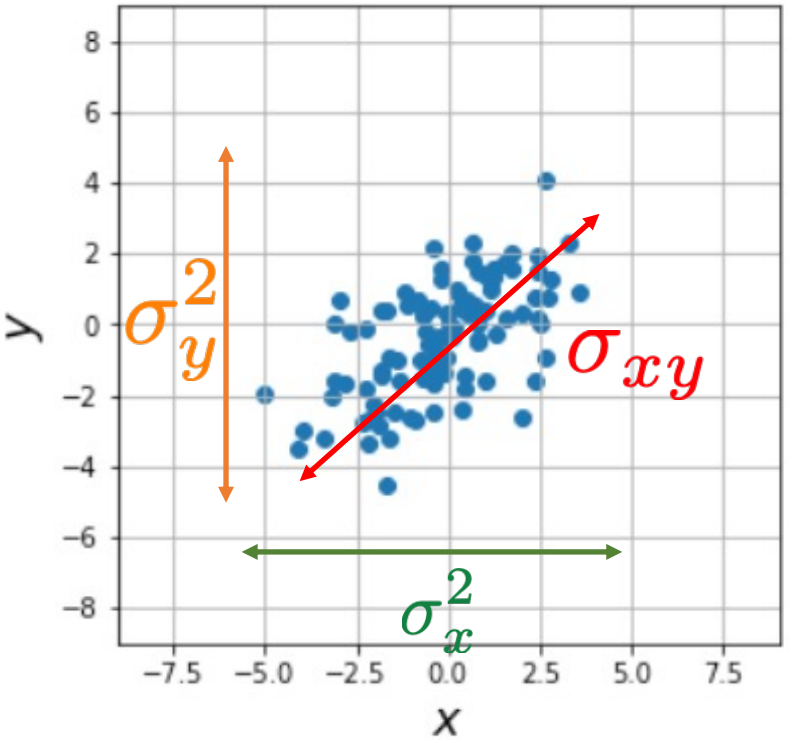
$$\Sigma_{xy} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$$

Note: One uses ρ_{XY} for a *population*

$$stde(r_{xy}) = \sqrt{\frac{1 - r^2}{N - 2}}$$

$$T = \frac{r_{xy}}{stde(r_{xy})} \equiv t(N - 2)$$



Covariance matrix: `numpy.cov(x, y, ddof=1)`
Pearson r: `numpy.corrcoef(x, y)`
Scipy (incl. p-val): `scipy.stats.pearsonr(x, y)`